

## KOKONAO TIMUKA, Indonesia

WMO: 977960

Lat: 4.5297S

Lon: 136.8925E

Elev: 3

StdP: 101.29

Time Zone: 9.00 (E09)

Period: 10-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 22.1       | 22.5 | 21.3                          | 15.9 | 24.0 | 21.6 | 16.3 | 24.1 | 5.9                   | 27.3 | 5.2 | 26.9 | 0.5                   | N/A  | 0.269 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 12            | 7.8                    | 33.3            | 26.6 | 32.8 | 26.5 | 32.3 | 26.5 | 27.4                | 31.6 | 27.1 | 31.4 | 26.8 | 31.0 | 3.0                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 26.2                            | 21.6 | 29.5 | 25.9 | 21.3 | 29.2 | 25.7 | 20.9 | 28.9 | 86.7          | 32.0 | 85.4 | 31.7 | 84.4 | 31.4 | 29.7           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 5.7           | 4.8  | 4.1 | DB                         | 21.4 | 34.6               | 0.5 | 0.4                                                | 21.0 | 34.9       | 20.8 | 35.1       | 20.5 | 35.4       | 20.2 | 35.7 | (4) |
| (5)               |      |     | WB                         | 20.8 | 28.6               | 0.6 | 0.6                                                | 20.4 | 29.0       | 20.1 | 29.4       | 19.8 | 29.8       | 19.3 | 30.2 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.1  | 27.8  | 27.7  | 27.8  | 27.6  | 27.2  | 26.2  | 25.5  | 25.6  | 26.4  | 27.3  | 27.8  | 27.9 | (6)  |
| (7)  |                                                                                  | DBStd       | 1.24  | 0.83  | 0.81  | 0.86  | 0.71  | 0.86  | 1.06  | 0.93  | 1.04  | 0.97  | 0.86  | 0.82  | 0.85 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD10.0     | 6225  | 552   | 496   | 551   | 528   | 533   | 485   | 481   | 483   | 492   | 536   | 533   | 554  | (10) |
| (11) |                                                                                  | CDD18.3     | 3183  | 294   | 262   | 293   | 278   | 275   | 235   | 223   | 225   | 242   | 277   | 283   | 296  | (11) |
| (12) |                                                                                  | CDH23.3     | 26268 | 2676  | 2343  | 2702  | 2493  | 2388  | 1654  | 1264  | 1323  | 1746  | 2373  | 2588  | 2717 | (12) |
| (13) | CDH26.7                                                                          | 8166        | 992   | 854   | 983   | 834   | 693   | 333   | 184   | 217   | 419   | 742   | 911   | 1002  | (13) |      |
| (14) | Wind                                                                             | WSAvg       | 1.1   | 1.2   | 1.2   | 1.2   | 1.1   | 1.0   | 1.1   | 1.2   | 1.2   | 1.1   | 1.1   | 1.1   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 5600  | 393   | 368   | 452   | 545   | 470   | 552   | 656   | 576   | 432   | 387   | 364   | 459  | (15) |
| (16) |                                                                                  | PrecMax     | 8061  | 778   | 683   | 788   | 827   | 861   | 1323  | 1483  | 1216  | 837   | 745   | 655   | 802  | (16) |
| (17) |                                                                                  | PrecMin     | 3253  | 175   | 165   | 201   | 288   | 227   | 189   | 136   | 8     | 33    | 64    | 158   | 205  | (17) |
| (18) |                                                                                  | PrecStd     | 1187  | 152   | 165   | 167   | 137   | 157   | 270   | 287   | 296   | 191   | 170   | 115   | 156  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 33.9  | 33.6  | 33.9  | 33.1  | 32.4  | 31.6  | 30.8  | 30.8  | 32.0  | 33.0  | 33.5  | 33.7 | (19) |
| (20) |                                                                                  |             | MCWB  | 26.7  | 26.6  | 26.5  | 26.3  | 26.5  | 26.3  | 26.0  | 25.6  | 26.6  | 26.5  | 26.9  | 26.8 | (20) |
| (21) |                                                                                  | 2%          | DB    | 33.0  | 32.8  | 33.0  | 32.2  | 31.6  | 30.6  | 29.5  | 29.7  | 30.9  | 32.0  | 32.7  | 32.9 | (21) |
| (22) |                                                                                  |             | MCWB  | 26.5  | 26.4  | 26.4  | 26.5  | 26.4  | 26.0  | 25.5  | 25.4  | 26.1  | 26.2  | 26.6  | 26.7 | (22) |
| (23) |                                                                                  | 5%          | DB    | 32.2  | 32.1  | 32.2  | 31.6  | 30.9  | 29.6  | 28.6  | 28.8  | 30.0  | 31.2  | 32.0  | 32.2 | (23) |
| (24) |                                                                                  |             | MCWB  | 26.5  | 26.3  | 26.4  | 26.4  | 26.2  | 25.8  | 25.2  | 25.1  | 25.6  | 26.1  | 26.4  | 26.6 | (24) |
| (25) |                                                                                  | 10%         | DB    | 31.4  | 31.2  | 31.3  | 30.8  | 30.2  | 28.7  | 27.7  | 27.9  | 29.1  | 30.4  | 31.2  | 31.4 | (25) |
| (26) |                                                                                  |             | MCWB  | 26.3  | 26.2  | 26.2  | 26.2  | 26.1  | 25.5  | 24.9  | 24.8  | 25.4  | 25.9  | 26.2  | 26.4 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 27.6  | 27.4  | 27.6  | 27.3  | 27.2  | 26.8  | 26.5  | 26.4  | 27.0  | 27.3  | 27.7  | 27.6 | (27) |
| (28) |                                                                                  |             | MADB  | 32.1  | 31.6  | 32.1  | 31.1  | 30.9  | 30.2  | 29.2  | 29.3  | 30.9  | 31.4  | 32.4  | 32.2 | (28) |
| (29) |                                                                                  | 2%          | WB    | 27.1  | 26.9  | 27.1  | 27.0  | 26.8  | 26.3  | 25.9  | 25.8  | 26.4  | 26.8  | 27.1  | 27.1 | (29) |
| (30) |                                                                                  |             | MADB  | 31.7  | 31.5  | 31.5  | 30.8  | 30.2  | 29.4  | 28.6  | 28.7  | 29.9  | 30.7  | 31.6  | 31.7 | (30) |
| (31) |                                                                                  | 5%          | WB    | 26.7  | 26.6  | 26.7  | 26.6  | 26.5  | 26.0  | 25.5  | 25.4  | 26.0  | 26.4  | 26.7  | 26.8 | (31) |
| (32) |                                                                                  |             | MADB  | 31.1  | 30.9  | 30.9  | 30.3  | 29.8  | 29.0  | 27.8  | 27.9  | 29.2  | 30.1  | 30.9  | 31.1 | (32) |
| (33) |                                                                                  | 10%         | WB    | 26.4  | 26.2  | 26.4  | 26.3  | 26.2  | 25.6  | 25.1  | 25.0  | 25.6  | 26.1  | 26.4  | 26.5 | (33) |
| (34) |                                                                                  |             | MADB  | 30.5  | 30.3  | 30.3  | 29.8  | 29.3  | 28.1  | 27.1  | 27.2  | 28.4  | 29.6  | 30.3  | 30.6 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 8.0   | 8.0   | 7.9   | 7.4   | 6.5   | 5.3   | 4.8   | 5.2   | 6.1   | 7.0   | 7.6   | 7.8  | (35) |
| (36) |                                                                                  |             | MCDBR | 9.0   | 8.9   | 9.0   | 8.2   | 7.5   | 6.6   | 6.1   | 6.4   | 7.4   | 8.0   | 8.6   | 8.8  | (36) |
| (37) |                                                                                  |             | MCWBR | 3.5   | 3.5   | 3.4   | 3.3   | 3.0   | 2.7   | 2.7   | 2.6   | 3.1   | 3.2   | 3.4   | 3.5  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 8.4   | 8.3   | 8.1   | 7.6   | 7.0   | 6.1   | 5.7   | 5.9   | 6.8   | 7.5   | 8.0   | 8.2  | (38) |
| (39) |                                                                                  |             | MCWBR | 3.6   | 3.6   | 3.4   | 3.3   | 3.0   | 2.7   | 2.7   | 2.7   | 3.1   | 3.3   | 3.4   | 3.5  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.406 | 0.403 | 0.399 | 0.400 | 0.415 | 0.418 | 0.417 | 0.436 | 0.443 | 0.480 | 0.457 | 0.424 | (40) |      |
| (41) |                                                                                  | taud        | 2.494 | 2.503 | 2.521 | 2.521 | 2.478 | 2.477 | 2.487 | 2.422 | 2.399 | 2.271 | 2.341 | 2.445 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 930   | 932   | 924   | 899   | 858   | 841   | 850   | 854   | 871   | 852   | 876   | 909   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 114   | 114   | 111   | 107   | 107   | 105   | 105   | 116   | 123   | 142   | 132   | 119   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 5.12  | 5.10  | 5.10  | 4.79  | 4.38  | 3.79  | 3.65  | 3.89  | 4.32  | 4.74  | 4.86  | 4.89  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.19  | 0.22  | 0.18  | 0.21  | 0.23  | 0.26  | 0.17  | 0.19  | 0.28  | 0.26  | 0.18  | 0.23  | (45) |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page